# VU Quiz Collection

## Total Quizzes: 10

**Courses: MTH603 - Numerical Analysis (Quiz No 02)**

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## Quiz 1 - MTH603 - Numerical Analysis (Quiz No 02)

**Question:**

Two matrices with the _____ characteristic polynomial need not be similar.

**Options:**

A. same

B. different

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## Quiz 2 - MTH603 - Numerical Analysis (Quiz No 02)

**Question:**

L e t [ A ] b e a 3 × 3 r e a l s y m m e t r i c m a t r i x w i t h | a 1 3 | b e t h e n u m e r i c a l l y l a r g e s t o f f " d i a g o n a
v a l u e o f ¸ c a n b e f o u n d b y L e t [ A ] b e a 3 × 3 r e a l s y m m e t r i c m a t r i x w i t h | a 1 3 | b e t h e n u m e r i c a
u s i n a g J a c o b i 2 s m e t h o d t h e v a l u e o f ¸ c a n b e f o u n d b y \ b e g i n { a r r a y } { l }
Let\,\left[ A \right]\,be\,a\,3 \times 3\,real\,symmetric\,matrix\,with\,\left| {{a_{13}}} \right|\,be\,the\,numerically\,\,l\arg
est\,off - diagonal\,element\,then\,\\u\sin g\,Jacobi's\,method\,the\,value\,of\,\theta \,can\,be\,found\,by
\end{array}

**Options:**

A. t a n 2 ¸ = 2 a 1 2 a 1 1 " a 2 2 t a n a 2 ¸ = 2 a 1 2 a 1 1 " a 2 2 \ t a n 2 \ t h e t a = \ f r a c { { 2 { a _ { 1 2 } } } } {

B. t a n 2 ¸ = 2 a 2 3 a 3 3 " a 2 2 t a n a 2 ¸ = 2 a 2 3 a 3 3 " a 2 2 \ t a n 2 \ t h e t a = \ f r a c { { 2 { a _ { 2 3 } } } } {

C. t a n 2 ¸ = 2 a 1 3 a 1 1 " a 3 3 t a n a 2 ¸ = 2 a 1 3 a 1 1 " a 3 3 \ t a n 2 \ t h e t a = \ f r a c { { 2 { a _ { 1 3 } } } } {

D. t a n 2 ¸ = 2 a 2 3 a 2 2 " a 3 3 t a n a 2 ¸ = 2 a 2 3 a 2 2 " a 3 3 \ t a n 2 \ t h e t a = \ f r a c { { 2 { a _ { 2 3 } } } } {

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## Quiz 3 - MTH603 - Numerical Analysis (Quiz No 02)

**Question:**

WhileusingtherelaxationmethodforfindingthesolutionofthebelowgivensystemwhichofthefollowingincrementwillbeintroC

duced?6x1" 2x2+3x3=1 " 2x1+7x2+2x3=5 x1+x2" 5x3=" 13 Whileusingtherelaxationmethodforfindi ebelowgivensystem,whichofthefollowingincrementwillbeintroduced?6x1" 2x2+3x3=1 " 2x1+7x2+ " 13 \begin{gathered}
While\,using\,the\,relaxation\,method\,for\,finding\,the\,solution\,of\,the\,below\,given\,system,\, \hfill \\
which\,of\,the\,following\,increment\,will\,be\,introduced? \hfill \\ \begin{array}{*{20}{l}}
{6{x_1} - 2{x_2} + 3{x_3} = 1} \\\
{ - 2{x_1} + 7{x_2} + 2{x_3} = 5} \\\
{{x_1} + {x_2} - 5{x_3} = - 13}
\end{array} \hfill \\\
\end{gathered}

**Options:**

A. dx2=R2a22dx2=R2a22d{x_2} = \frac{{{R_2}}}{{{a_{22}}}}

B. dx3=R3a33dx3=R3a33d{x_3} = \frac{{{R_3}}}{{{a_{33}}}}

C. dx1=R1a11dx1=R1a11d{x_1} = \frac{{{R_1}}}{{{a_{11}}}}

D. dx1=R1a22dx1=R1a22d{x_1} = \frac{{{R_1}}}{{{a_{22}}}}

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## Quiz 4 - MTH603 - Numerical Analysis (Quiz No 02)

**Question:**

The Gauss-Seidel method is applicable to strictly diagonally dominant or symmetric _____ definite matrices A.

**Options:**

A. negative

B. positive

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## Quiz 5 - MTH603 - Numerical Analysis (Quiz No 02)

**Question:**

While using Jacobi method for the matrix
A =#¡#£#¢ 2 0 0 0 2 1 0 1 2#¤#¦#¥ A = [ 2 0 0 0 2 1 0 1 2 ] A = \left[ {\begin{array}{*{20}{c}}

2&0&0\\0&2&1\\0&1&2
\end{array}} \right]
and 'theta ¸ =pi/4' the orthogonal matrix S1 will be given as

**Options:**

A. S 1 =#¡#£#¢#¢ c o s À 4 s i n À 4 0" s i n À 4 c o s À 4 0 0 0 1#¤#¦#¥#¥ S 1 = [ c o s a À 4" s i n a À 4 0 s i n a À 4 c
{\begin{array}{*{20}{c}}
{\cos \frac{\pi }{4}}&{ - \sin \frac{\pi }{4}}&0\\{\sin \frac{\pi }{4}}&{\cos \frac{\pi }{4}}&0\\0&0&1
\end{array}} \right]

B. S 1 =#¡#£#¢#¢ c o s À 4 0 s i n À 4 0 1 0" s i n À 4 0 c o s À 4#¤#¦#¥#¥ S 1 = [ c o s a À 4 0" s i n a À 4 0 1 0 s i n a
{\begin{array}{*{20}{c}}
{\cos \frac{\pi }{4}}&0&{ - \sin \frac{\pi }{4}}\\0&1&0\\{\sin \frac{\pi }{4}}&0&{\cos \frac{\pi }{4}}
\end{array}} \right]

C. S 1 =#¡#£#¢#¢ 1 0 0 0 c o s À 4 s i n À 4 0" s i n À 4 c o s À 4#¤#¦#¥#¥ S 1 = [ 1 0 0 0 c o s a À 4" s i n a À 4 0 s i n
{\begin{array}{*{20}{c}}
1&0&0\\0&{\cos \frac{\pi }{4}}&{ - \sin \frac{\pi }{4}}\\0&{\sin \frac{\pi }{4}}&{\cos \frac{\pi }{4}}
\end{array}} \right]

D. S 1 =#¡#£#¢#¢ c o s À 4 0 0 0" s i n À 4 0 0 0 1#¤#¦#¥#¥ S 1 = [ c o s a À 4 0 0 0" s i n a À 4 0 0 0 1 ] { S _ 1 } = \
{\cos \frac{\pi }{4}}&0&0\\0&{ - \sin \frac{\pi }{4}}&0\\0&0&1
\end{array}} \right]

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## Quiz 6 - MTH603 - Numerical Analysis (Quiz No 02)

**Question:**

While using Jacobi method for the matrix
A =#¡#£#¢ 1 3 –" 2 –" 3 –" 3 2 2 –" 2 1#¤#¦#¥ A = [ 1 3 2 3 3 2 2 2 1 ] A = \left[ {\begin{array}{*{20}{c}}
1&{\sqrt 3 }&{\sqrt 2 }\\{\sqrt 3 }&3&2\\{\sqrt 2 }&2&1
\end{array}} \right]
the value of 'theta ¸' can be found as

**Options:**

A. tan2¸ = 2 a 1 3 a 1 1" a 3 3 t a n a 2¸ = 2 a 1 3 a 1 1" a 3 3 \tan 2\theta = \frac{{2{a_{13}}}}{

B. tan2¸ = 2 a 2 3 a 2 2" a 3 3 t a n a 2¸ = 2 a 2 3 a 2 2" a 3 3 \tan 2\theta = \frac{{2{a_{23}}}}{

C. tan2¸ = 2 a 1 2 a 1 1" a 2 2 t a n a 2¸ = 2 a 1 2 a 1 1" a 2 2 \tan 2\theta = \frac{{2{a_{12}}}}{

D. tan2¸ = 2 a 1 3 a 3 3" a 1 1 t a n a 2¸ = 2 a 1 3 a 3 3" a 1 1 \tan 2\theta = \frac{{2{a_{13}}}}{

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## Quiz 7 - MTH603 - Numerical Analysis (Quiz No 02)

**Question:**

Direct methods can be more rapid than iterative algorithms.

**Options:**

A. TRUE

B. FALSE

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## Quiz 8 - MTH603 - Numerical Analysis (Quiz No 02)

**Question:**

By using determinants, we can easily check that the solution of the given system of linear equation exits and it is unique.

**Options:**

A. FALSE

B. TRUE

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## Quiz 9 - MTH603 - Numerical Analysis (Quiz No 02)

**Question:**

While using Jacobi method for the matrix
A =#¡#£#¢ 1 1 / 4 1 / 2 1 / 4 1 / 3 1 / 4 1 / 2 1 / 4 1 / 5#¤#¦#¥ A = [ 1 1 / 4 1 / 2 1 / 4 1 / 3 1 / 4 1 / 2 1 / 4 1 / 5 ] A = \left[
1&{1/4}&{1/2}\\{1/4}&{1/3}&{1/4}\\{1/2}&{1/4}&{1/5}
\end{array}} \right]
and ‘theta ¸ =0.4480, the orthogonal matrix S1 will be given by

**Options:**

A. S1=#¡#£#¢cos0.4480000" sin0.448000001#¤#¦#¥S1=[cosa0.4480000" sina0.44800

{\begin{array}{*{20}{c}}
{\cos 0.4480}&0&0\\0&{ - \sin 0.4480}&0\\0&0&1
\end{array}} \right]

B. S1=#¡#£#¢cos0.44800sin0.44800010" sin0.44800cos0.4480#¤#¦#¥S1=[cosa0.44800" sina0.4480010s
{\begin{array}{*{20}{c}}
{\cos 0.4480}&0&{ - \sin 0.4480}\\0&1&0\\{\sin 0.4480}&0&{\cos 0.4480}
\end{array}} \right]

C. S1=#¡#£#¢cos0.44800sin0.44800" sin0.44800cos0.448000001#¤#¦#¥S1=[cosa0.4480" sina0.44800sina0
{\begin{array}{*{20}{c}}
{\cos 0.4480}&{ - \sin 0.4480}&0\\{\sin 0.4480}&{\cos 0.4480}&0\\0&0&1
\end{array}} \right]

D. S1=#¡#£#¢10000cos0.44800sin0.44800" sin0.44800cos0.4480#¤#¦#¥S1=[10000cosa0.4480" sina0.44800
{\begin{array}{*{20}{c}}
1&0&0\\0&{\cos 0.4480}&{ - \sin 0.4480}\\0&{\sin 0.4480}&{\cos 0.4480}
\end{array}} \right]

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## Quiz 10 - MTH603 - Numerical Analysis (Quiz No 02)

**Question:**

The dominant or principal eigenvector of a matrix is an eigenvector corresponding to the eigenvalue of largest magnitude (for real numbers, largest absolute value) of that matrix.

**Options:**

A. TRUE

B. FALSE

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